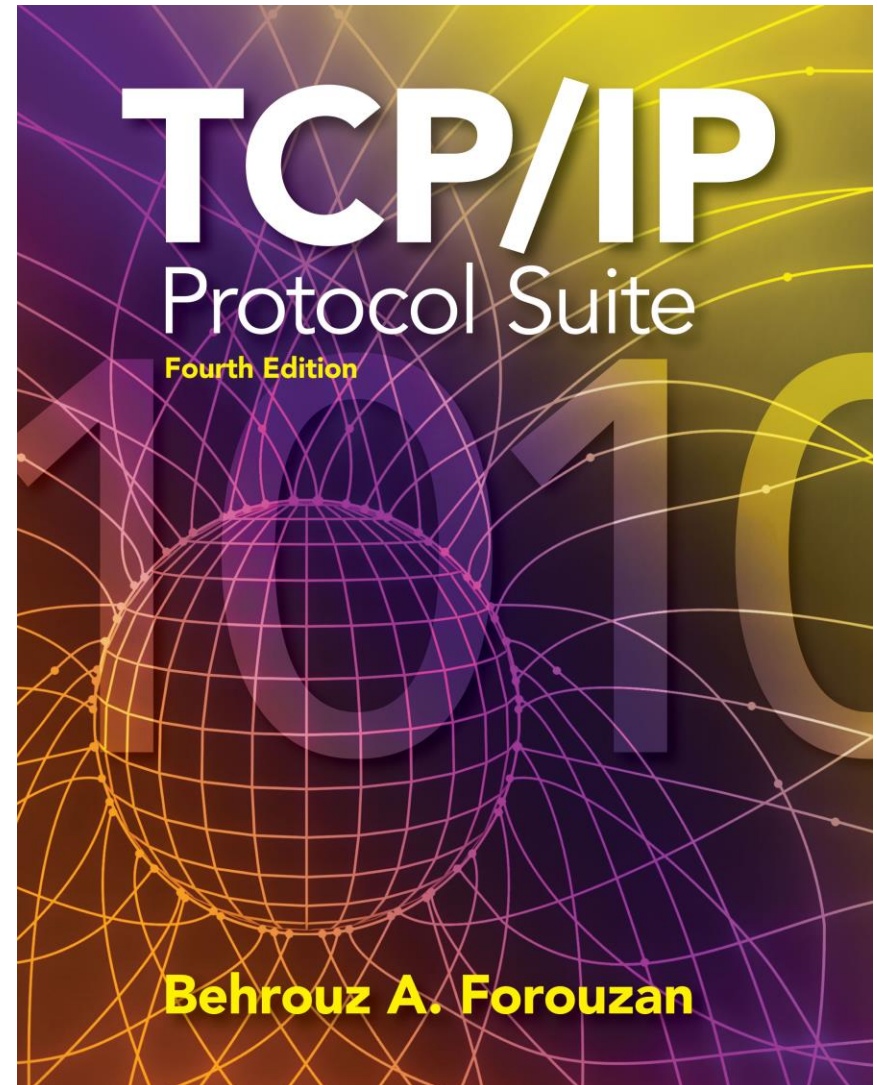


Chapter 1

Introduction



OBJECTIVES:

- ☐ **To give a brief history of the Internet.**
- ☐ **To give the definition of the two often-used terms in the discussion of the Internet: protocol and standard.**
- ☐ **To categorize standard organizations involved in the Internet and give a brief discussion of each.**
- ☐ **To define Internet Standards and explain the mechanism through which these standards are developed.**
- ☐ **To discuss the Internet administration and give a brief description of each branch.**

Chapter Outline

1.1 A Brief History

1.2 Protocols and Standards

1.3 Standards Organizations

1.4 Internet Standards

1.5 Internet Administration

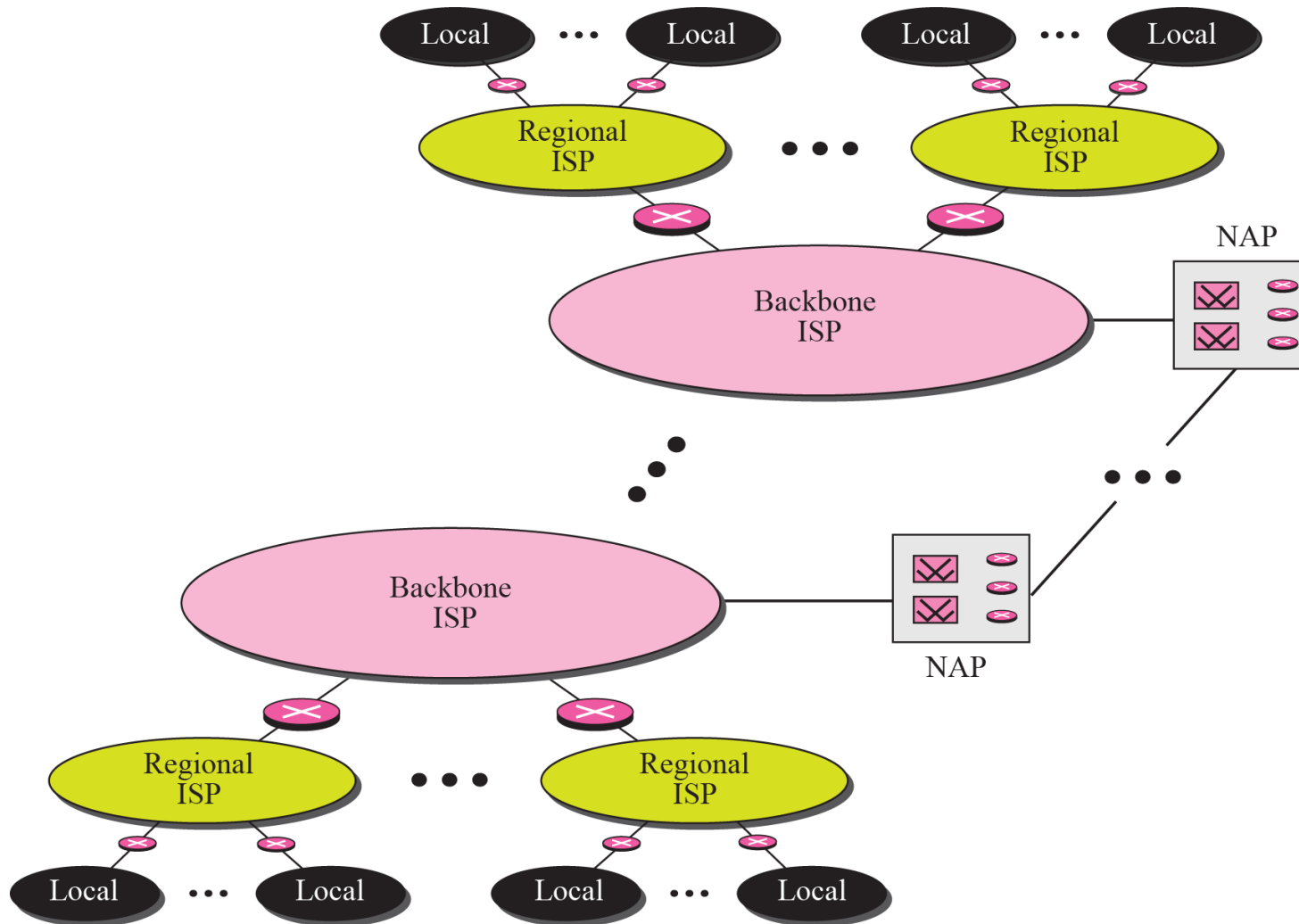
1-1 A BRIEF HISTORY

A network is a group of connected, communicating devices such as computers and printers. An *internet* is two or more networks that can communicate with each other. The most notable internet is called *the internet*, composed of hundreds of thousands of interconnected networks. Private individuals as well as various organizations such as government agencies, schools, research facilities, corporations, and libraries in more than 100 countries use the Internet.

Topics Discussed in the Section

- ✓ **ARPANET**
- ✓ **Birth of the Internet**
- ✓ **TCP/IP**
- ✓ **MILNET**
- ✓ **CSNET**
- ✓ **NSFNET**
- ✓ **ANSNET**
- ✓ **The Internet Today**
- ✓ **World Wide Web**
- ✓ **Growth of the Internet**

Figure 1.1 *Internet today*



1-2 PROTOCOLS AND STANDARDS

In this section, we define two widely used terms: protocols and standards. First, we define *protocol*, which is synonymous with “rule.” Then we discuss *standards*, which are agreed-upon rules.

Topics Discussed in the Section

- ✓ **Protocols**
- ✓ **Standards**

1-3 STANDARDS ORGANIZATION

Standards are developed through the cooperation of standards creation committees, forums, and government regulatory agencies.

Topics Discussed in the Section

- ✓ **Standards Creation Committees**
- ✓ **Forums**
- ✓ **Regulatory Agencies**

1-4 INTERNET STANDARDS

An Internet standard is a thoroughly tested specification that is useful to and adhered to by those who work with the Internet. It is a formalized regulation that must be followed. There is a strict procedure by which a specification attains Internet standard status. A specification begins as an Internet draft. An Internet draft is a working document with no official status and a six-month lifetime.

Topics Discussed in the Section

- ✓ **Maturity Levels**
- ✓ **Requirement Levels**

Figure 1.2 *Maturity levels of an RFC*

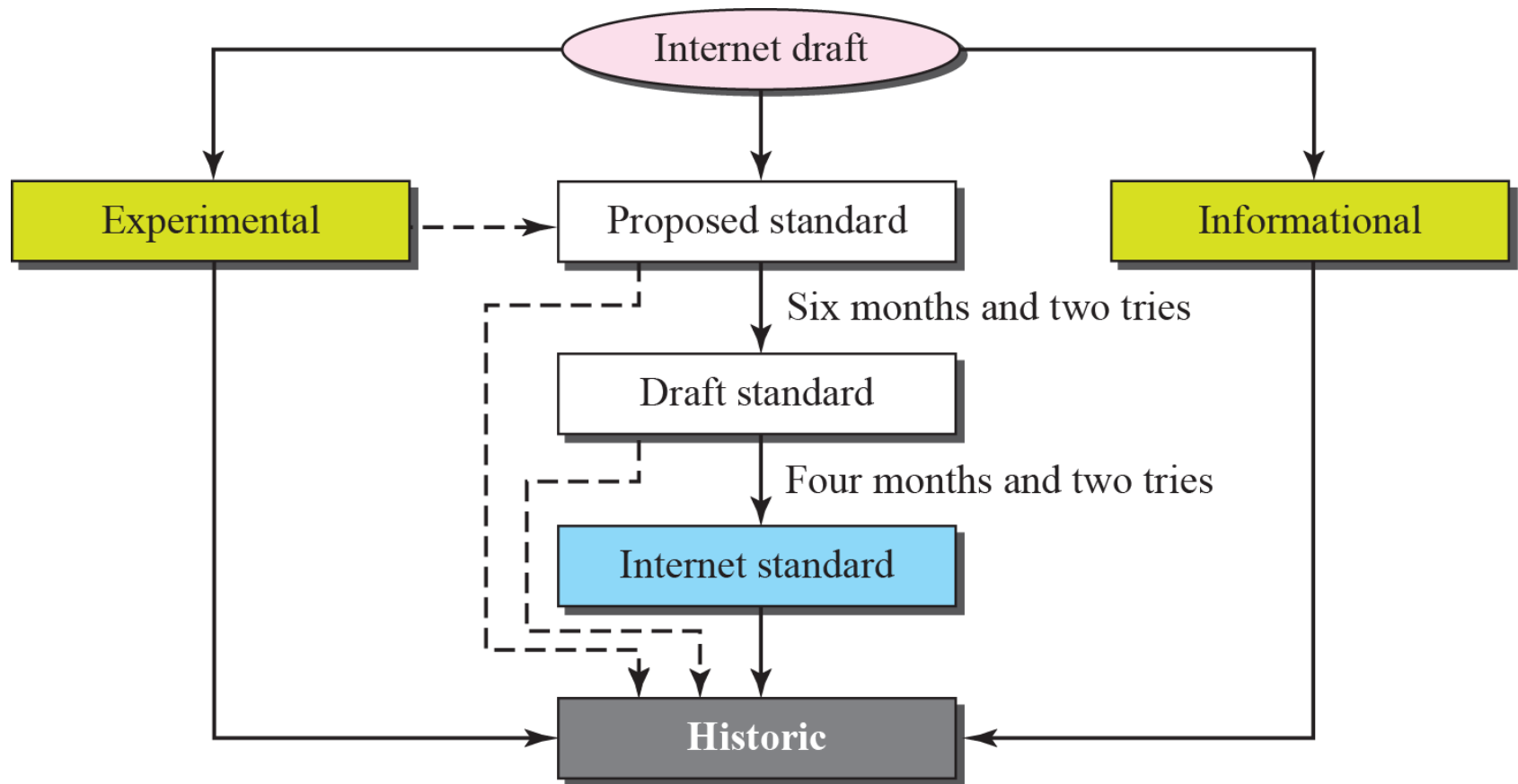
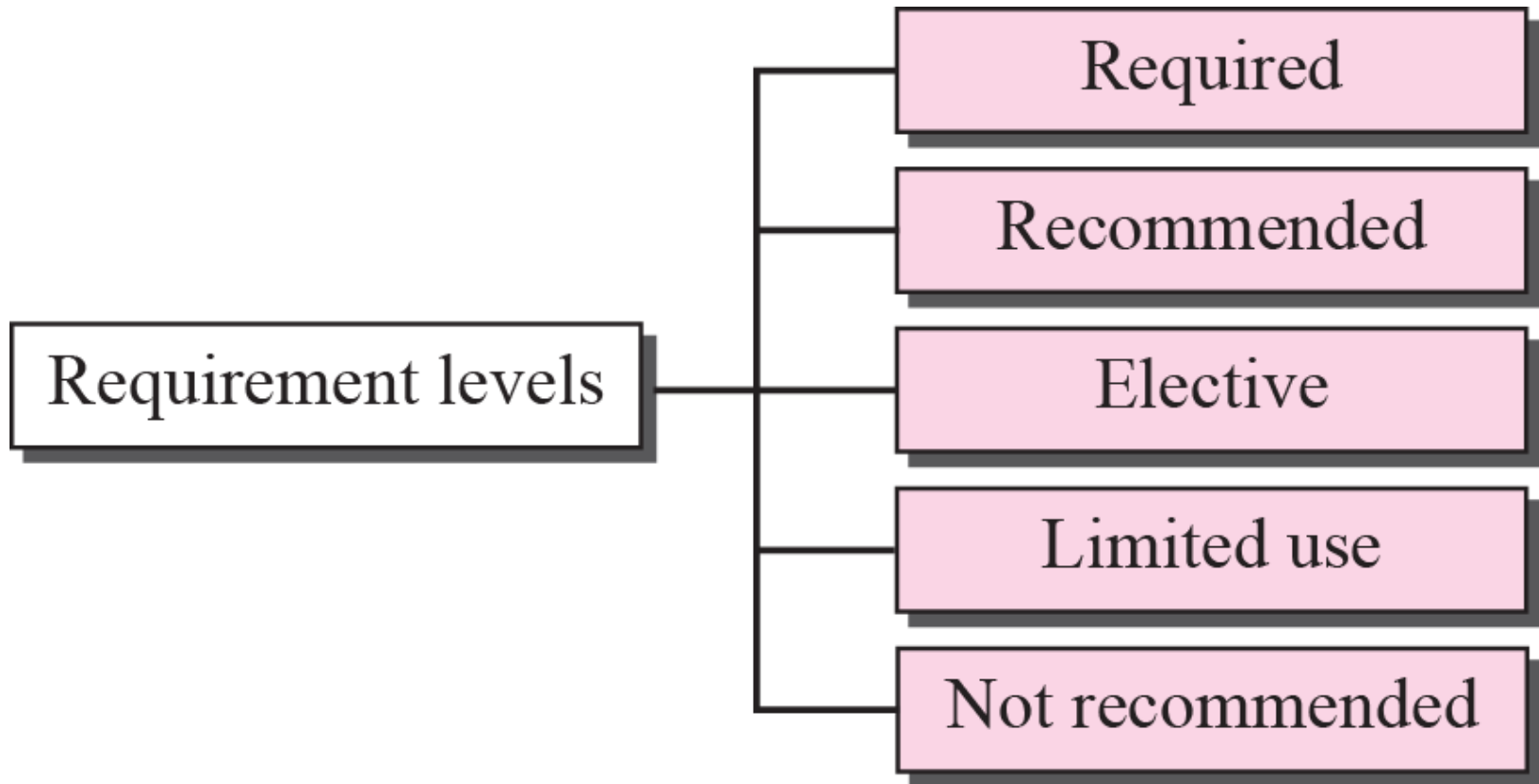


Figure 1.3 *Requirement levels of an RFC*





Note

*RFCs can be found at
<http://www.rfc-editor.org>.*

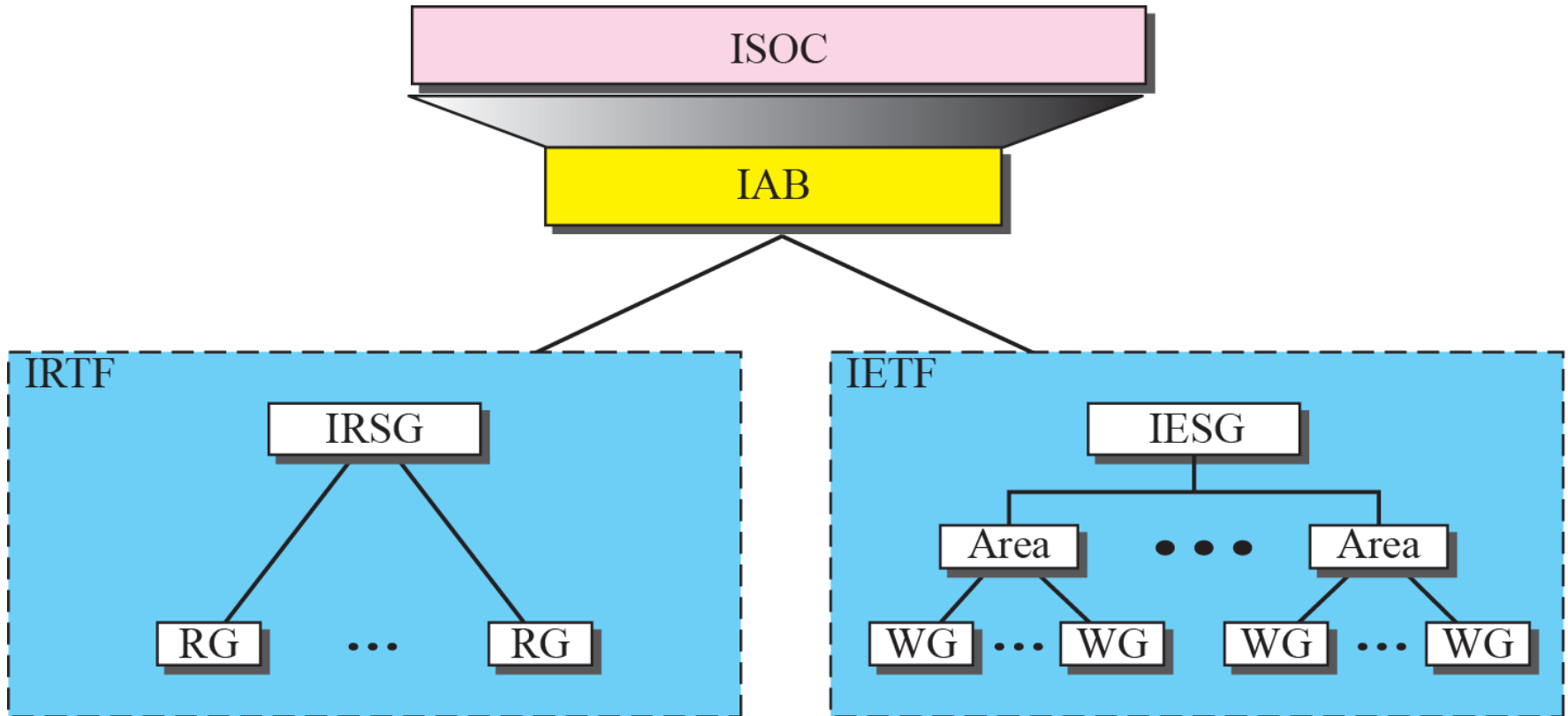
1-5 INTERNET ADMINISTRATION

The Internet, with its roots primarily in the research domain, has evolved and gained a broader user base with significant commercial activity. Various groups that coordinate Internet issues have guided this growth and development. Appendix G gives the addresses, e-mail addresses, and telephone numbers for some of these groups. Figure 1.4 shows the general organization of Internet administration.

Topics Discussed in the Section

- ✓ **Internet Society (ISOC)**
- ✓ **Internet Architecture Board (IAB)**
- ✓ **Internet Research Task Force (IRTF)**
- ✓ **Internet Assigned Number Authority (IANA)**
- ✓ **Internet Corporation for Names and Numbers (ICANN)**
- ✓ **Network Information Center (NIC)**

Figure 1.4 *Internet administration*





Note

The addresses and websites for Internet organizations can be found in Appendix G.